



Genetic control of source–sink relationships in grain sorghum

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Abstract

Main conclusion QTL hotspots identified for selected source–sink-related traits provide the opportunity for pyramiding favorable alleles for improving sorghum productivity under diverse environments.

Abstract A sorghum bi-parental mapping population was evaluated under six different environments at Hays and Manhattan, Kansas, USA, in 2016 and 2017, to identify genomic regions controlling source–sink relationships. The population consisted of 210 recombinant inbred lines developed from US elite post-flowering drought susceptible (RTx430) and a known post-flowering drought tolerant cultivar (SC35). Selected physiological traits related to source (effective quantum yield of photosystem II and chlorophyll index), sink (grain yield per panicle) and panicle neck diameter were recorded during grain filling. The results showed strong phenotypic and genotypic association between panicle neck diameter and grain yield per panicle during mid-grain filling and at maturity. Multiple QTL model revealed 5–12 including 2–5 major QTL for each trait. Among them 3, 7 and 8 QTL for quantum yield, panicle neck diameter and chlorophyll index, respectively, have not been identified previously in sorghum. Phenotypic variation explained by QTL identified across target traits ranged between 5.5 and 25.4%. Panicle neck diameter and grain yield per panicle were positively associated, indicating the possibility of targeting common co-localized QTL to improve both traits simultaneously through marker-assisted selection. Three major QTL hotspots, controlling multiple traits were identified on chromosome 1 (52.23–61.18 Mb), 2 (2.52–11.43 Mb) and 3 (1.32–3.95 Mb). The identified genomic regions and underlying candidate genes can be utilized in pyramiding favorable alleles for improving source–sink relationships in sorghum under diverse environments.

Keywords Multi-environment · Panicle neck diameter · Quantitative trait loci · Sorghum · Source–sink dynamics

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Introduction

Sorghum [*Sorghum bicolor* (L.) Moench], originating from tropical and subtropical regions of Africa, is adapted to arid, semi-arid, sub-humid and humid environments (Smith and Frederiksen 2000). Sorghum is considered to be one of the hardiest crops to heat and drought stress conditions (Doggett 1988; Blum 2004; Smith and Frederiksen 2000). Eighty percent of sorghum in the world is grown under dry, rainfed environments (Assefa et al. 2010). Global average yield of sorghum is only 1.2 t ha⁻¹ (Jordan et al. 2012). Low productivity in sorghum compared with other grain crops is attributed primarily to harsh growing environments and minimal inputs, resulting in exposure to many abiotic stresses and nutritional deficiencies. However, previous studies have demonstrated sorghum's high production potential when grown under non-stress conditions. A study in Australia analyzed 23 field trials from 12 locations over 5 years, exposed to varying moisture conditions reported mean sorghum yield ranging between 2.3 and 10.5 t ha⁻¹ (Jordan et al. 2012). Another study using hybrid grain sorghum performance trials at Kansas from 1957 to 2008 documented sorghum yields of up to 9.3 t ha⁻¹ under irrigated conditions (Assefa and Staggenborg 2010). Hence, investigating traits that contribute towards sorghum's adaptability and productivity spanning diverse environments is an interesting research objective.

Grain yield is primarily determined by a series of physiological processes that result in production and transport of photoassimilates from the source (leaf) to the sink (grain). Leaf chlorophyll and photosynthesis are key physiological traits for photoassimilate production, while panicle (sink strength) plays an important role in transporting assimilates from the source to the sink. Hence, understanding genetic architecture of these traits and their interrelationship under a diverse set of environments could help in improving sorghum yield. Identifying genomic regions that control these associated traits, followed by validation of marker–trait associations provide the foundation for marker-assisted selection (MAS) and breeding for yield improvement (Collard et al. 2005).

Previous studies have identified several Quantitative Trait Loci (QTL) for chlorophyll index in sorghum and some of them co-localized with stay-green loci under post-flowering drought stress (Xu et al. 2000; Gelli et al. 2016; Sukumaran et al. 2016). The positive association between chlorophyll index or stay-green loci with yield under drought stress has been documented and are considered as important genomic regions for improving drought adaptation in sorghum (Vadez et al. 2013; Borrell et al. 2014). Identifying new genomic regions associated with chlorophyll index would help towards pyramiding favorable

alleles for further enhancing chlorophyll concentrations and photosynthetic efficiency under water-limited environments. On the other hand, limited QTL for chlorophyll fluorescence has been reported in sorghum exposed to variable environments under field conditions. Recently, chilling tolerance studies under controlled environment have identified genomic regions associated with chlorophyll fluorescence and gas exchange traits during early vegetative growth stages using GWAS (Fiedler et al. 2016; Ortiz et al. 2017). Except for a study by Sukumaran et al. (2016), there are no efforts to map QTL for photosynthetic efficiency during reproductive stages in field conditions in sorghum. Hence, identifying novel QTL associated with chlorophyll fluorescence under diverse field environments would be useful to understand and further improve photosynthetic efficiency in sorghum.

Several QTL related to panicle architecture in sorghum such as panicle length, panicle diameter, panicle width, panicle neck length, and panicle exertion have been identified (Zou et al. 2012; Fakrudin et al. 2013; Reddy et al. 2013; Hmon et al. 2014; Zhang et al. 2015). Positive correlation between grain yield with panicle architecture traits including panicle length, panicle width, and panicle diameter has been reported in sorghum (Hmon et al. 2013). However, none of the previous studies have investigated panicle neck diameter and QTL that control this trait. A positive relationship between panicle neck diameter and yield and related parameters have been reported in other crops such as rice (Liu et al. 2008), even under water-limited conditions (Quinones et al. 2017). However, such a relationship has not been established in sorghum. Panicle neck is a major route that connects the source with the sink and, hence, plays a key role in photoassimilate transport from leaf to grain. Identifying QTL associated with panicle neck diameter during different panicle maturity stages will provide additional opportunities to optimize source and sink dynamics and enhance productivity in sorghum grown under diverse environments.

QTL related to grain yield and other yield-related parameters have been documented in different genetic backgrounds in sorghum. Previous studies have been conducted under natural environments in the field (Rami et al. 1998; Zou et al. 2012; Fakrudin et al. 2013; Reddy et al. 2013, 2014; Gelli et al. 2016; Tao et al. 2018, 2020, 2021), with only a few studies including both well-watered and drought stress environments (Tuinstra et al. 1997; Sabadin et al. 2012; Phuong et al. 2013). Among previous studies there are no attempts to map QTL for grain yield during mid-grain filling, exposed to diverse environments and moisture conditions including full irrigation throughout the crop duration, partial irrigation and no irrigation under field conditions.

The bi-parental Recombinant Inbred Lines (RIL) population (SC35 x RTx430) used in this study is one of the 10 RIL families of a sorghum Nested Association Mapping

(NAM) population, (Yu et al. 2013; Bouchet et al. 2017; Perumal et al. 2021). SC35 is an important stay-green sorghum line converted from durra landrace IS12555 (Stephens et al. 1967). IS12555 is a stay-green source that originated in Ethiopia and its derivative line, BTx642 (formerly known as B35) has been used in many previous studies as post-flowering drought tolerant stay-green check line to identify stay-green and chlorophyll index loci (Tuinstra et al. 1997; Crasta et al. 1999; Subudhi et al. 2000; Xu et al. 2000; Sanchez et al. 2002). SC35 has comparatively low yield potential than RTx430 (Mahama et al. 2014). RTx430 is a widely adapted elite line with high yield potential and has been commonly used in many breeding programs in the United States (Miller 1984; Yu et al. 2013). Although RTx430 is not a stay-green line, it is also a non-senescence genotype which maintains leaf greenness throughout its growth (Miller 1984). Hence, it has similar chlorophyll and photosynthetic efficiency under non-water limiting conditions (Crasta et al. 1999; Mahama et al. 2014). However, it is susceptible to post-flowering drought stress; hence, its chlorophyll index decreases under water-limited conditions (Crasta et al. 1999).

The objectives of this study were to (1) identify QTL associated with source, sink and panicle neck diameter traits in different environments with varying soil moisture; (2) establish an overview of the relationships between the putative loci associated with source, sink and panicle neck diameter traits identified in the present study and previously mapped genes/QTL; and (3) determine phenotypic and genotypic relationship among selected physiological traits that determine yield. We hypothesized that (1) large variation in grain yield in the population will lead to identifying major QTL during mid grain filling and at maturity, (2) variation in panicle neck diameter will lead to identification of novel QTL, and (3) chlorophyll index and effective quantum yield of photosystem II QTL will co-localize with known stay-green loci.

Materials and methods

Plant material and experimental design

The SC35 x RTx430 mapping population consisting of 210 RILs, along with their parents were tested in two environments [(irrigated throughout season—(environment 1) and irrigated until heading (environment 2)] at Agricultural Research Center, Hays (ARCH), Kansas, USA in 2016. The same set of RILs was planted at four environments in 2017. Three experiments [irrigated throughout season (environment 3), irrigated until heading (environment 4) and natural dryland environment (environment 5)] were conducted at ARCH, Kansas; and one experiment was carried out under natural environment (environment 6) at Manhattan, Kansas.

In total, the population was tested under six different environments with considerable range in temperature and soil moisture conditions (Table 1). The population was planted at Hays on June 2 in 2016 and on May 25 in 2017 and in Manhattan on June 15, 2017. All experiments were set up in a randomized complete block design (RCBD) with two replications. The experiment had one row plot in 2016 and 2 rows plot in all experiments in 2017. Each row was 3.6 m long and seeds were planted with inter-row spacing of 0.75 m. Air temperature and precipitation were recorded using 1 or 2 WatchDog data loggers (1000 Series Micro Station, Spectrum Technologies, Aurora, IL, USA) in each experimental location. Volumetric soil moisture content was measured in environment 3 and environment 5, at about 10 cm below soil surface using a soil moisture sensor per environment (WaterScout SM 100 moisture sensor, Spectrum Technologies, Aurora, IL, USA).

Phenotypic observations

The population was evaluated for physiological traits related to source [(effective quantum yield of Photosystem II (QY) and chlorophyll index (Ch)], sink (grain yield per panicle) and panicle neck diameter. Effective quantum yield of photosystem II in the light adapted state was measured using FluorPen (FluorPen FP 100, Photon System Instruments, Ltd., Brno, Czech Republic) (Šebela et al. 2018). Chlorophyll index was measured using Dualex (Dualex 4 Scientific, Force-A, Orsay, France) in 2016 (Chiluwal et al. 2018) and SPAD meter (SPAD-502 Plus, Konica Minolta) in 2017. Both these measurements were recorded on the flag leaf between 9:30 am and 3:30 pm on bright sunny days, over three–four different time-points during the grain-filling phase in each location. Grain weight per panicle (g panicle^{-1}) and respective panicle neck diameter were recorded at two time-points, i.e., mid grain-filling stage and physiological maturity leading to grain yield per panicle at mid grain-filling stage (MGY), final grain yield per panicle (FGY), panicle neck diameter at mid grain-filling stage (MND), and final panicle neck diameter (FND).

In each plot, three uniform heads from the main tiller were tagged at the time of anthesis using different color ribbons, i.e., red, green and yellow to mark 1st, 2nd and 3rd head, respectively. Due to differences in phenology within the population (Fig. 1a), each tagged panicle was monitored at 1- to 2-day intervals to record the exact date of complete flowering. Mid grain-filling stage was considered at exactly 15 days after complete flowering to overcome any differences in flowering phenology in the population and the parents. Panicles were manually harvested to record grain yield per panicle at mid grain-filling stage and their respective panicle neck diameter was recorded using digital vernier calipers (VWR® Digital Calipers, USA) just below the panicle

Table 1 Soil and environmental details from different locations of the study

Env	Treatment	Location	Year	Soil characteristics						Days > 35 °C ^a	Moisture % ^b
				pH	Mehlich P ppm	K ppm	NH ₄ -N ppm	NH ₃ -N ppm	Sand %	Silt %	Clay %
1	Irrigated throughout season	Hays	2016	8.28	26.73	408.00	2.35	27.55	20.67	60.67	18.67
2	Irrigated until heading	Hays	2016	8.19	32.43	468.67	2.04	33.33	21.33	60.67	18.00
3	Irrigated throughout season	Hays	2017	8.01	70.67	466.00	2.05	17.47	21.33	62.00	16.67
4	Irrigated until heading	Hays	2017	8.02	25.67	435.00	1.84	24.56	20.00	62.67	17.33
5	Natural dryland environment	Hays	2017	7.68	16.60	557.67	3.02	18.14	18.00	48.00	34.00
6	Natural environment	Manhattan	2017	5.49	27.77	259.67	22.52	51.25	12.00	57.33	30.67

^aNumber of days when daily maximum temperature exceeded 35 °C^bAverage volumetric soil moisture content 10 cm below the soil surface between July and September

node. After physiological maturity, three uniform panicles from main tillers were harvested from each plot to record final grain yield per panicle and their respective panicle neck diameter were recorded as mentioned above. Panicles were dried at 40 °C for a week before threshing. Plant height was also recorded in environments 3, 4 and 5 to identify well-known plant height QTL for validating the correctness of the genotypic data and QTL analysis from this study. Plant height was recorded from the base of the plant to the tip of the panicle at the time of final harvest.

QTL analysis

Existing linkage map (Bouchet et al. 2017) for the bi-parental population generated from a cross between SC35 and RTx430 was used for QTL analysis. Briefly, the map incorporates 23,400 single nucleotide polymorphism (SNP) markers and spans 1293 cM in length. Multiple QTL (MQM) analysis was conducted on normally distributed data based on Haley–Knott regression using stepwiseqtl function as implemented in R/qtl package (Broman et al. 2003) for each trait separately. LOD threshold for each QTL analysis was determined separately based on 1,000 permutations on a reduced genotype data generated using reduce2grid function. This step reduced the number of markers to approximately 1293 and was only used for the calculation of LOD. By default, the MQM model in R/qtl automatically tests for presence of epistasis interaction. The % PVE (Phenotypic Variation Explained), average additive effects and 1.5 LOD interval were estimated from the best fit QTL model obtained using built in fitqtl function in R/qtl. QTL explaining at least 10% phenotypic variation was considered as a major QTL. The archive maintained at Sorghum QTL Atlas (Mace et al. 2019) if available or the QTL data from the published papers was used to study co-localization with previously reported QTL in sorghum. The co-localization of QTL for four studied traits was considered as QTL hotspot. QTL were named as letter q followed by an acronym for the trait and chromosome number (for, e.g., *qMGY1.2* denote second QTL on chromosome 1 for grain yield at mid grain-filling stage).

Statistical analysis

Phenotypic correlation coefficients between the traits were calculated for each environment separately, between the environments and also average across all environments. Analysis of variance was conducted for each environment using generalized linear mixed model in R and means were separated using lsmeans. Broad-sense heritability was calculated from estimated variance components of genotype (V_g) and prediction error variance (PEV) as $H = 1 - (PEV/V_g)$ (Clark et al. 2012; Poudel et al. 2019) which is equivalent

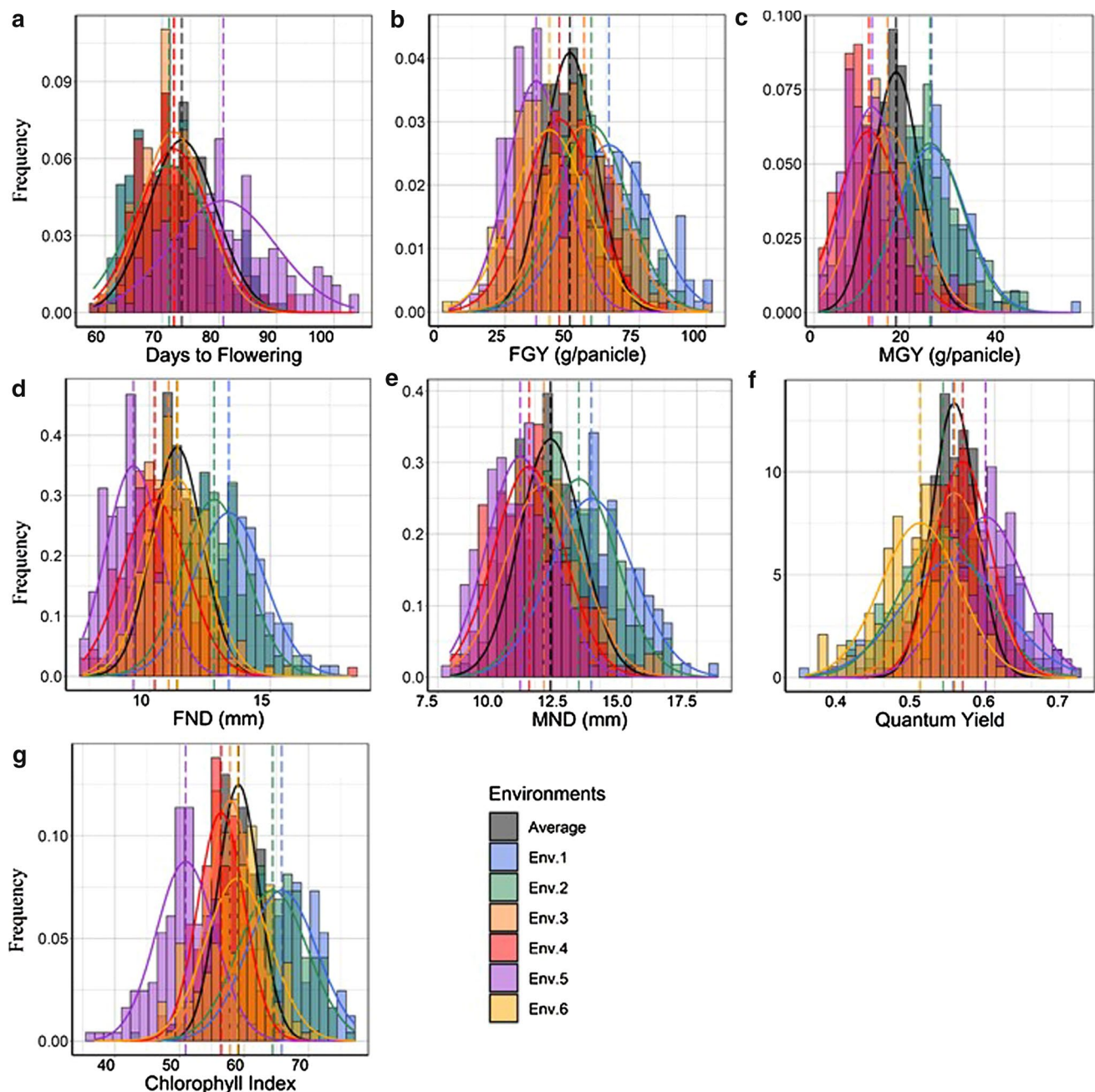


Fig. 1 Histogram showing frequency distribution of the traits in each and averaged across all the environments. **a** Days to 50% flowering; **b** final grain yield (g panicle^{-1}); **c** grain yield at mid grain-filling stage

(g panicle^{-1}); **d** final panicle neck diameter (mm); **e** panicle neck diameter at mid grain-filling stage (mm); **f** effective quantum yield of photosystem II, **g** chlorophyll index

to genotype-mean heritability. The phenotypic values used for QTL analysis across environment were based on random effects model where genotype (G), environment (E), $G \times E$ and replication were considered random factors. In case of individual environment analysis, phenotype values for grain yield per panicle and neck diameter were calculated based on random effect model with G and replication as random factors.

Results

Environmental conditions

Considerable variation in soil texture, soil nutrient status, temperature, precipitation and soil moisture were observed among the environments (Table 1, Supplementary Fig. 1). Cumulative precipitation throughout the crop growing

season (May to October) was highest in Manhattan 2017 (445 mm) followed by Hays 2016 (395 mm) and Hays 2017 (356 mm) (Supplementary Fig. 1d). Soil moisture sensors recorded 19.4, 18.6 and 3.3% average volumetric soil moisture in July, August and September, respectively, in the natural dryland environment 5, and 28.3, 29.1 and 25% in July, August and September in the irrigated environment 3. Total number of days with daily maximum temperature exceeding 35 °C throughout the crop growing season was 34, 33 and 19 days in Hays 2016, Hays 2017 and Manhattan 2017, respectively (Table 1).

Phenotypic trait variation

All the studied traits showed normal distribution with skewness < 1.0 suggesting all traits had polygenic inheritance (Fig. 1). Analysis of variance (ANOVA) revealed significant effect of genotype, environment and genotype $\times$ environment interaction ($G \times E$) for all the traits (Supplementary Table 1), indicating the need to conduct QTL analysis separately for each environment.

The phenotypic differences for both the parents, and RILs averaged across all the environments is presented in Table 2, and for each environment is presented in Supplementary Table 2. Across environments, parents varied significantly only in panicle neck diameter but not with other traits (Table 2). SC35 recorded significantly higher

final panicle neck diameter and panicle neck diameter at mid grain-filling stage in all the environments than RTx430 except in environments 5 and 6 (Supplementary Tables 2a–f). Differences between the parents in other traits were also observed in some of the environments (Supplementary Table 2a–f). RTx430 had a significantly higher final grain yield per panicle and grain yield per panicle at mid grain-filling stage in environment 1 (Supplementary Table 2a). Significant differences in effective quantum yield of photosystem II between the parents were observed in environments 1 and 2 (Supplementary Table 2a, b). Similarly, significant differences in chlorophyll index between the parents was observed in environment 6 (Supplementary Table 2f). The RILs grown under natural dryland environment in Hays (environment 5) recorded lower chlorophyll index, grain yield per panicle (except at mid grain-filling stage in environment 4) and panicle neck diameter at both stages, but a higher effective quantum yield of photosystem II than other environments (Fig. 1, Supplementary Table 2g). RILs showed transgressive segregation resulting in a wide variation for all the studied traits in each environment and averaged across all the environments (Table 2, Supplementary Table 2a–2f). Moderate to high broad sense heritability (h^2) was observed for all studied traits ranging between 0.58 (effective quantum yield of photosystem II) and 0.85 (final panicle neck diameter) (Table 2).

Table 2 Summary statistics for all studied traits in RILs and parents averaged across all environments

Traits	RIL Population				
	Min	Max	Mean	SEM	h^2
FGY	22.19	74.2	49.09	0.68	0.80
MGY	7.48	34.83	17.31	0.33	0.72
FND	7.48	14.04	11.16	0.07	0.85
MND	9.06	15.02	11.88	0.08	0.84
QY	0.47	0.63	0.55	0.01	0.58
Ch	48.85	64.74	58.07	0.21	0.78
Traits	Parent ^a				
	SC35	SEM	RTx430	SEM	
FGY	53.61 ^a	3.39	54.95 ^a		4.48
MGY	17.70 ^a	1.09	19.95 ^a		1.96
FND	11.91 ^a	0.71	10.55 ^b		0.49
MND	12.62 ^a	0.62	10.64 ^b		0.43
QY	0.54 ^a	0.01	0.54 ^a		0.01
Ch	54.01 ^a	1.27	56.31 ^a		1.49

FGY final grain yield (g panicle⁻¹), MGY grain yield at mid grain-filling stage (g panicle⁻¹), FND final panicle neck diameter (mm), MND panicle neck diameter at mid grain-filling stage (mm), QY effective quantum yield of photosystem II, Ch chlorophyll index, SEM standard error of mean, h^2 broad sense heritability

^aNumber followed by different letter within a row for the same trait are significantly different at $p=0.05$

Relationship between source-related traits with others

Chlorophyll index had significant but weak association with panicle neck diameter at both stages averaged across all environments ($r=0.20\text{--}0.26$) (Fig. 2b), and within each environment ($r=0.14\text{--}0.28$) (Fig. 2a). Chlorophyll index was weakly associated with effective quantum yield of photosystem II ($r=0.17\text{--}0.18$, in environments 4 and 6, respectively) but in environments 1, 2, 3 and 5, they were not significantly associated (Fig. 2a). Chlorophyll index was significantly correlated with final grain yield per panicle ($r=0.14\text{--}0.24$) in 2 out of 6 environments, while negatively associated with grain yield per panicle at mid grain-filling stage ($r=-0.14$ and -0.16) in environments 3 and 4. Effective quantum yield of photosystem II was not found to be consistently

associated with any other traits although both significant positive and negative weak association (-0.14 to 0.21) with other traits were observed in few combinations (Fig. 2a).

Relationship between grain yield and panicle neck diameter

Phenotypic correlation between yield per panicle and panicle neck diameter in both mid-grain filling and at maturity were significant across all environments ($r=0.36\text{--}0.52$) (Fig. 2b), and in all two-way combinations in each environment ($r=0.15\text{--}0.56$) (Fig. 2a). Significant and strong positive correlations were observed in panicle neck diameter between mid-grain filling and final stage in each environment ($r=0.54\text{--}0.71$) (Fig. 2a) and averaged phenotypic value across all environments ($r=0.84$) (Fig. 2b). Similarly, grain

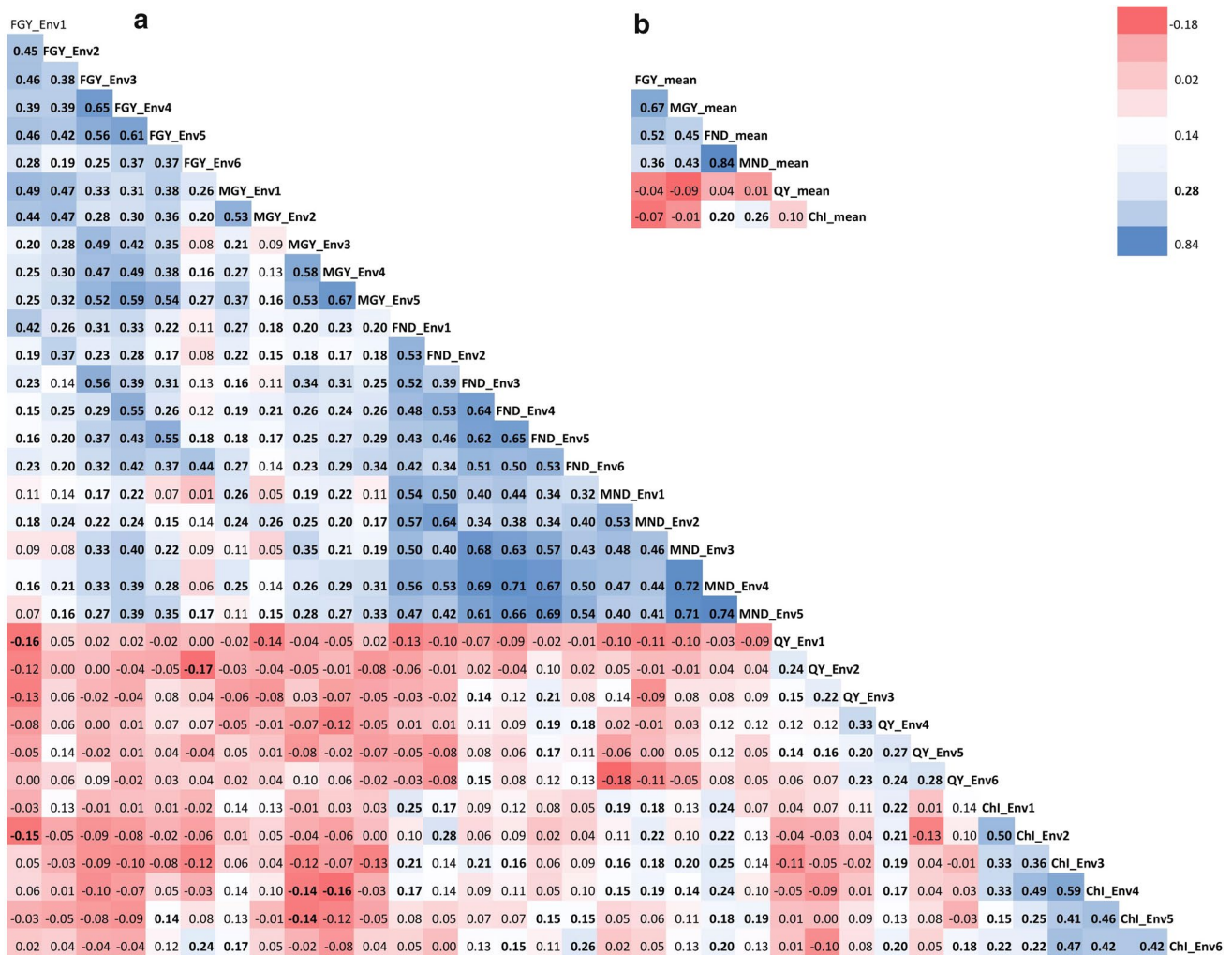


Fig. 2 Phenotypic correlation between the studied traits in each, between (a) and averaged across all environments (b). The Pearson correlation coefficient value (r) with bold font represents significant correlation at the 0.05 probability level. Env environment, FGY final

grain yield (g panicle^{-1}), MGY grain yield at mid grain-filling stage (g panicle^{-1}), FND final panicle neck diameter (mm), MND panicle neck diameter at mid grain-filling stage (mm), QY effective quantum yield of photosystem II, Chl chlorophyll index

yield per panicle between the two stages was also strongly associated in each environment (0.47–0.54) (Fig. 2a) and averaged across all environments ($r=0.67$) (Fig. 2b).

Correlation between environments

Final grain yield per panicle ($r=0.19$ – 0.65), final panicle neck diameter ($r=0.34$ – 0.65), and panicle neck diameter at mid grain-filling stage ($r=0.40$ – 0.74), were significantly associated between the environments (Fig. 2a). Significant positive association was observed in most environment combinations for grain yield per panicle at mid grain-filling stage ($r=0.16$ – 0.67) except between environment 2 with environments 3 and 4 (Fig. 2a), and similarly for effective quantum yield of photosystem II ($r=0.14$ – 0.33) except between environments 1 and 2 with environments 4 and 6 (Fig. 2a). Chlorophyll index measured using Dualex in environments 1 and 2, and using SPAD meter in environments 3, 4, 5 and 6 were also significantly associated ($r=0.15$ – 0.33) between the environments (Fig. 2a). The association in chlorophyll index between the environments measured with the same device was higher (on average $r=0.46$ with SPAD meter and $r=0.50$ with Dualex) than different devices (on average $r=0.29$) (Fig. 2a).

QTL mapping

Grain yield

Multiple QTL mapping (MQM) identified six QTL, including 3 major QTL for grain yield at mid-grain filling on chromosomes 1, 3, 4, 6 and 10 (Table 3 and Supplementary Table 3). Two major QTL on chromosome 1 were detected in multiple environments. These were located at opposite sides of the centromere (23.6–32.1 cM and 94.2–131.8 cM; Fig. 3). The other major QTL were identified at the distal end of chromosome 3 (Table 3). PVE by these QTL ranged from 5.6 to 19.3% (Supplementary Table 3). Parents SC35 and RTx430 possessed favorable alleles for 4 and 2 QTL, respectively (Supplementary Table 3).

Five QTL explaining 6.6–14.5% phenotypic variation for final grain yield were detected (Supplementary Table 4), of which three each on chromosomes 1, 2, and 3 were deemed as major QTL (Table 3). Favorable alleles for all major QTL were contributed by SC35, while RTx430 possessed favorable allele for remaining minor QTL on chromosomes 5 and 6 (Supplementary Table 4). Two major QTL each on chromosomes 1 and 3 were detected in multiple environments (Table 3).

Two major QTL identified for final grain yield and grain yield at mid grain-filling stage were found to be co-localized (Fig. 3). These include QTL at chromosome 1 (94–132 cM) and chromosome 3 (0–18 cM). One minor QTL for final

grain yield at chromosome 6 was co-localized with QTL for grain yield at mid grain-filling stage at a position between 28 and 81 cM (Fig. 3).

Panicle neck diameter

Five QTL for panicle neck diameter at mid grain-filling stage were identified on chromosomes 1, 2 and 9 (Supplementary Table 5). Phenotypic variation explained by these QTL ranged from 6.2 to 21.6%, with 2 of them deemed as major QTL (Supplementary Table 5 and Table 3). SC35 and RTx430 possessed favorable alleles in 4 and 1 QTL, respectively (Supplementary Table 5). Both major QTL for panicle neck diameter at mid grain-filling stage were located on chromosome 2 without overlapping, with one of them (13.1–36 cM) detected in 3 environments and in the QTL analysis with the average phenotypic value across all environments (Table 3 and Fig. 4). A significant epistatic interaction was observed between a QTL on chromosome 2 (*qMND2.2*) and chromosome 9 (*qMND9.1*) (Supplementary Table 5).

Eight QTL, including 3 major QTL explaining 5.5–25.4% variation in final panicle neck diameter were detected on chromosomes 1, 2, 3, 6, 7 and 9 (Table 3 and Supplementary Table 6). All these QTL had positive additive effect (Supplementary Table 6), indicating the exclusive source of higher final panicle neck diameter in the population is contributed by SC35. Three major QTL on chromosomes 1, 2, and 3 were detected in multiple environments (Table 3). A major QTL (*qFND2.1*) on chromosome 2 at position 10–50 cM was detected in 5 environments and in the QTL analysis with average phenotypic value across all environments with PVE up to 25.4% (Table 3). The other two major QTL on chromosomes 1 and 3 were detected in multiple environments and contributed up to 14.4 and 12.5% phenotypic variation, respectively (Table 3). A significant epistatic interaction was observed between a QTL on chromosome 2 (*qFND2.1*) and chromosome 6 (*qFND6.1*) (Supplementary Table 6).

Parent SC35 having higher panicle neck diameter possessed favorable alleles for all major QTL in both stages. One major QTL and two minor QTL for final panicle neck diameter and panicle neck diameter at mid grain-filling stage were co-located, respectively (Fig. 3). Major QTL for final panicle neck diameter (*qFND2.1*) and panicle neck diameter at mid grain-filling stage (*qMND2.1*) on chromosome 2 (10–50 cM) were co-localized, each of which explained more than 21% of phenotypic variation (Fig. 3, Supplementary Table 5 and Supplementary Table 6). Similarly, one QTL on chromosome 1 (*qFND1.1* with *qMND1.1*), and one QTL on chromosome 9 (*qFND9.1* with *qMND9.2*) were found to co-localize for both final panicle neck diameter and panicle neck diameter at mid grain-filling stage (Fig. 3).

Table 3 List of major QTL detected for all traits

QTL ^a	Env ^b	Peak SNP	Position (cM)	QTL interval (cM)	LOD	Additive effect ^c	PVE	Previously mapped QTL ^d	Previously mapped trait description
Grain yield at mid grain-filling stage—MGY									
<i>qMGY1.1</i>	1, 2	S1_10857585	25.9	23.6–32.1	7.25	1.46	19.26	<i>QGNMB1.1^d</i> , <i>QGNMB1.2^d</i> , <i>QGYLD1.10^e</i> , <i>QGYLD1.11^e</i> , <i>QGYLD1.15^e</i> , <i>QGWGT1.3^f</i> , <i>QGWGT1.4^g</i> , <i>QGWGT1.25^h</i> , <i>qGS1.4ⁱ</i> , <i>qFHGS1.4^j</i>	Grain number, grain weight, grain yield, grain size
<i>qMGY1.2</i>	3, 5, all	S1_56951880	118	94.2–131.8	4.21	0.99	10.24	<i>QGYLD1.18^k</i> , <i>QGYLD1.21^k</i> , <i>QGYLD1.22^k</i> , <i>QGYLD1.23^k</i> , <i>QGYLD1.3^k</i> , <i>QGYLD1.13^e</i> , <i>QGYLD1.7^l</i> , <i>QGYLD1.8^l</i> , <i>QGWGT1.9^m</i> , <i>QGYLD1.4ⁿ</i> , <i>QGWGT1.8^o</i> , <i>QGWGT1.7^o</i> , <i>QGYLD1.25^f</i> , <i>QGYLD1.5^q</i> , <i>QGYLD1.6^r</i> , <i>qGS1.4ⁱ</i> , <i>qFHGS1.8^j</i> , <i>qFHGS1.9^j</i> , <i>qFHGS1.10^j</i>	Grain weight, grain yield, grain size
<i>qMGY3.1</i>	1, all	S3_2724553	12.5	0.7–18.1	5.84	1.09	10.52	<i>QGYLD3.23^s</i> , <i>QGWGT3.10^r</i> , <i>QGYLD3.1^m</i> , <i>QGYLD3.2^m</i> , <i>qFHGS3.2^j</i>	Grain weight, grain yield, grain size
Final grain yield—FGY									
<i>qFGY1.1</i>	2, 4, all	S1_55243671	107.3	102.9–132.1	5.20	2.39	12.13	<i>QGYLD1.18^k</i> , <i>QGYLD1.21^k</i> , <i>QGYLD1.22^k</i> , <i>QGYLD1.23^k</i> , <i>QGYLD1.30^k</i> , <i>QGYLD1.13^e</i> , <i>QGWGT1.9^m</i> , <i>QGYLD1.5^q</i> , <i>QGYLD1.7^l</i> , <i>QGYLD1.8^l</i> , <i>QGYLD1.6^r</i> , <i>qGS1.4ⁱ</i> , <i>qFHGS1.9^j</i> , <i>qFHGS1.10^j</i>	Grain weight, grain yield, grain size
<i>qFGY2.1</i>	4, all	S2_6795698	35.9	28–43.1	4.43	2.75	10.15	<i>QRPRT2.1^h</i> , <i>QPNUM2.1^u</i> , <i>QGYLD2.10^f</i> , <i>QGYLD2.9^v</i> , <i>qFHGS2.3^j</i>	Grain number, panicle number, grain yield, grain size
<i>qFGY3.1</i>	2, 5, all	S3_1325223	3.9	0.3–18.8	7.32	2.96	14.54	<i>QGYLD3.23^s</i> , <i>QGWGT3.10^r</i> , <i>QGYLD3.1^m</i> , <i>QGYLD3.2^m</i> , <i>QGYLD3.11^w</i> , <i>qGS3.1ⁱ</i> , <i>qFHGS3.1^j</i>	Grain weight, grain yield, grain size
Panicle neck diameter at mid grain-filling stage—MND									
<i>qMND2.1</i>	3, 4, 5, all	S2_3340593	15.5	13.1–36	12.81	0.46	21.62	None	None
<i>qMND2.2</i>	5, all	S2_13011922	54.4	36.5–73.6	9.44	0.23	15.22	None	None
Final panicle neck diameter—FND									
<i>qFND1.1</i>	4, 5, all	S1_60442493	130	116.6–130.5	7.9	0.23	14.35	None	None
<i>qFND2.1</i>	1, 2, 3, 4, 5, all	S2_5638382	30.2	10.6–50.7	15.90	0.42	25.35	None	None
<i>qFND3.2</i>	2, 4, 5, all	S3_60961434	109.1	101.4–126.4	7.12	0.25	12.50	None	None
Effective quantum yield of photosystem II—QY									

Table 3 (continued)

QTL ^a	Env ^b	Peak SNP	Position (cM)	QTL interval (cM)	LOD	Additive effect ^c	PVE	Previously mapped QTL ^d	Previously mapped trait description
<i>qQY2.2</i>	6.3	S2_70457570	119.7	115.2–132.3	5.76	0.03	12.19	<i>QCHLF2.27^x</i> , <i>QCHLF 2.29^x</i> , <i>QCHLF 2.30^x</i> , <i>QPS12.2^y</i>	Chlorophyll fluorescence, Photosynthetic efficiency
<i>qQY6.1</i>	all	S6_43089978	47.7	45.6–48.2	8.14	0.01	17.44	None	None
<i>qQY7.1</i>	6.3	S7_58871733	88.7	86.9–91.6	7.53	0.04	16.39	None	None
<i>qQY8.1</i>	all	S8_5349585	36.4	34.9–37.8	5.38	0.01	11.11	None	None
Chlorophyll index – Ch									
<i>qCh2.2</i>	2.1, 2.2, 2.3	S2_3340593	15.5	14.1–20.7	9.31	4.72	16.23	<i>QCHLC2.18^x</i> , <i>QCHLC 2.21^x</i>	Chlorophyll index
<i>qCh5.1</i>	6.2	S5_7508163	40.6	25.3–44.9	4.97	–2.56	14.08	None	None
<i>qCh7.1</i>	6.1	S7_58876466	88.9	72.1–90.9	4.18	2.06	10.94	<i>QCHLC7.15^x</i> , <i>QCHLC7.17^k</i> , <i>QCHLC7.20^k</i>	Chlorophyll index
<i>qCh8.1</i>	2.1, 3.2, all	S8_52215596	89.8	77.8–99.5	7.00	–0.83	11.52	<i>QCHLC8.1^x</i> , <i>QCHLC 8.12^z</i> , <i>QCHLC 8.13^x</i> , <i>QCHLC 8.14^z</i>	Chlorophyll index
<i>qCh10.1</i>	2.1, 2.3, 3.2, 6.1, all	S10_10822734	49.5	36.8–58.9	7.31	–1.09	12.10	<i>QCHLC10.1^z</i> , <i>QCHLC10.7^z</i> , <i>QCHLC10.8^z</i> , <i>QCHLC10.9^z</i> , <i>QCHLC10.10^z</i> , <i>QCHLC10.2^{a+}</i>	Chlorophyll index

All—average phenotypic value across all tested environments

Env environment, Chr chromosome, LOD logarithm of odds, PVE percent phenotypic variation explained

^aThe QTL name is designated as a combination of the trait under study followed by chromosome and QTL index. The QTL were indexed same if at least the one end of the QTL interval overlaps within the same chromosome

^bDigit after decimal represent time of the measurement, i.e., Env 6.3—third measurement in environment 6

^cAdditive effect—positive values indicate contribution by SC35 and negative values by RTx430

^dQTL name as listed in Sorghum QTL Atlas (Mace et al. 2018) if available or in the published paper and associated references; ^xReddy et al. (2013); ^ySukumaran et al. (2016); ^zGuindo et al. (2019); ^gSrinivas et al. (2009); ^hSpagnolli et al. (2016); ⁱTao et al. (2020); ^jTao et al. (2021); ^kGelli et al. (2016); ^mShehzad and Okuno (2015); ⁿKapanigowda et al. (2014); ^oPaterson et al. (1995); ^pFeltus et al. (2006); ^qShiringani et al. (2010); ^rZhang et al. (2015); ^sSabadin et al. (2012); ^tMocoeur et al. (2015); ^uSakhi et al. (2013); ^vRitter et al. (2008); ^wPhuong et al. (2013); ^xFiedler et al. (2016); ^yOrtiz et al. (2017); ^zFiedler et al. (2014); ^{a+}Reddy et al. (2014)

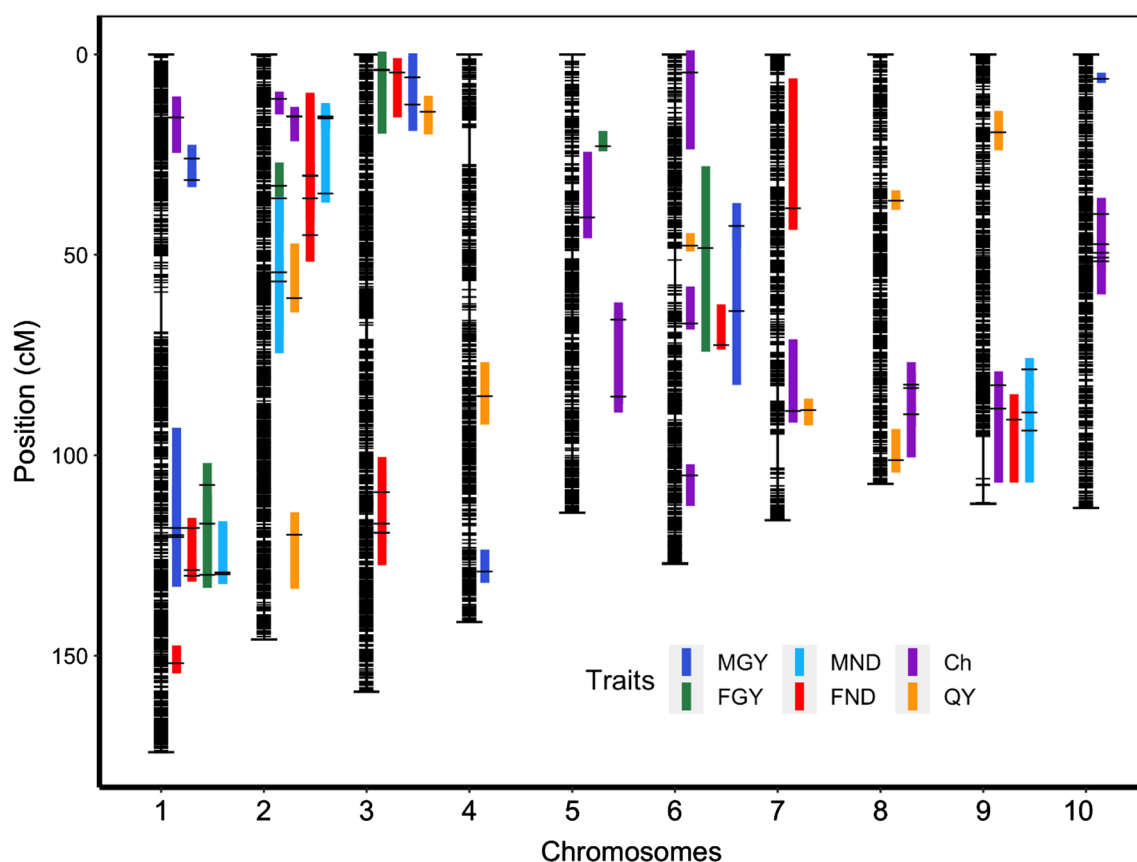


Fig. 3 Position of detected QTL for all studied traits. *MGY* grain yield at mid grain-filling stage (g panicle^{-1}), *FGY* final grain yield (g panicle^{-1}), *MND* panicle neck diameter at mid grain-filling stage

(mm), *FND* final panicle neck diameter (mm), *Ch* chlorophyll index, *QY* effective quantum yield of photosystem II

Effective quantum yield of photosystem II

Nine QTL (major—4, minor—5) for effective quantum yield of photosystem II were detected on chromosomes 2, 3, 4, 6, 7, 8 and 9 (Table 3 and Supplementary Table 7). Phenotypic variation explained by these QTL ranged from 7.8 to 17.4% (Supplementary Table 7). SC35 and RTx430 contributed favorable alleles for 4 and 5 QTL, respectively (Supplementary Table 7). A significant epistatic interaction was observed between a QTL on chromosome 6 (*qQY6.1*) and chromosome 8 (*qQY8.1*) (Supplementary Table 7). None of these QTL were detected in more than one environment, and also at multiple time periods in any environment.

Chlorophyll index

Twelve QTL for chlorophyll index were detected on all chromosomes except 3 and 4, with PVE ranging from 6.0 to 16.4% (Supplementary Table 8). SC35 and RTx430 possessed favorable alleles for 7 and 5 QTL, respectively (Supplementary Table 8). Five of them were major QTL, with

one each on chromosomes 8 and 10 detected in multiple environments (Table 3 and Fig. 3).

Plant height

Eight QTL, including 2 major QTL for plant height were detected on chromosomes 1, 2, 3, 4, 5 and 6 (Supplementary Table 9). Phenotypic variation explained by these QTL ranged from 6 to 18% (Supplementary Table 9). Both SC35 and RTx430 possessed favorable alleles for 4 QTL (Supplementary Table 9). Two major QTL each on chromosomes 2 and 6 were detected in all environments and in the QTL analysis with average phenotypic value across all environments (Supplementary Table 9). The major QTL for plant height on chromosome 6 was identified in three environments and in the QTL analysis with the average phenotypic value across all environments which contributed up to 18% phenotypic variation. The QTL were co-localized with well-known dwarfing locus *Dw2* in sorghum (Higgins et al. 2014; Bouchet et al. 2017; Hilley et al. 2017), validating the accuracy of the genotypic data and supporting the QTL analysis.

Major QTL hotspots

Three QTL hotspots for multiple traits were identified one each on chromosomes 1, 2 and 3. The region between 94.2 and 132.1 cM (52.2–61.2 Mb) on chromosome 1 contained major QTL for final grain yield (*qFGY1.1*), grain yield at mid grain-filling stage (*qMGY1.2*), final panicle neck diameter (*qFND1.1*), and a minor QTL for panicle neck diameter at mid grain-filling stage (*qMND1.1*) (Fig. 3). The region between 10.3 and 50.7 cM (2.5–11.4 Mb) on chromosome 2 had major QTL for final grain yield (*qFGY2.1*), final panicle neck diameter (*qFND2.1*), panicle neck diameter at mid grain-filling stage (*qMND2.1*) and chlorophyll index (*qCh2.2*). Similarly, the region between 0.3 and 19 cM (1.3–3.9 Mb) on chromosome 3 contained major QTL for final grain yield (*qFGY3.1*), grain yield at mid grain-filling stage (*qMGY3.1*), and minor QTL for final panicle neck diameter (*qFND3.1*) and effective quantum yield of photosystem II (*qQY3.1*) (Fig. 3).

Discussion

Based on the available literature, this is the first study in sorghum that has attempted to map QTL for source, sink and panicle neck diameter-related traits during different growth stages under different environment and moisture conditions, including full irrigation throughout the crop duration, partial irrigation and no irrigation under field conditions. The study detected major QTL for each traits including some novel QTL for panicle neck diameter, effective quantum yield and chlorophyll index.

Population tested under diverse environments

Experimental sites in both years received some rainfall in all the months throughout the growing season (Supplementary Fig. 1d); hence, the population grown under environments 2, 4 and 6 was not exposed to the anticipated level of water-deficit conditions. However, population grown under natural dryland environment in Hays (environment 5) experienced water-limited conditions during its terminal stage as no additional irrigation was provided during the entire growing season and precipitation was less in Hays compared with Manhattan in 2017 (Supplementary Fig. 1d). Due to lower soil moisture content (see Table 1), population grown in environment 5 recorded the lowest grain yield per panicle and panicle neck diameter during both final and at mid grain-filling stages among the environments (Fig. 1, Supplementary Table 2g) providing additional evidence of moderate terminal water-deficit stress compared with other environments. Temperature above 32 °C during the reproductive stage affects sorghum

yield as even short episodes of temperature beyond the critical threshold can cause significant grain yield reduction (Prasad et al. 2015; Tack et al. 2017). Differences in total number of days when daily maximum temperature exceeded 35 °C during the crop growing season (Table 1) suggest that plants grown in Hays (Environments 1–5) experienced significantly more hotter days compared to Manhattan (Environment 6).

Novel QTL for panicle neck diameter identified but not grain yield

Major QTL detected for final grain yield (*qFGY1.1*) and grain yield at mid grain-filling stage (*qMGY1.2*) on chromosome 1; final grain yield (*qFGY3.1*) and grain yield at mid grain-filling stage (*qMGY3.1*) on chromosome 3; final grain yield (*qFGY2.1*) on chromosome 2 and grain yield at mid grain-filling stage on chromosome 1 (*qMGY1.1*) have all been identified previously under different experimental conditions (Table 3, refer to associated references in the legend). The major QTL identified on chromosome 1 (*qMGY1.1*) was also reported by Sukumaran et al. (2016), indicating the robustness of this QTL for grain yield as it was consistently detected in multiple environments and accounted for 16% of the phenotypic variation in their study. Our findings provide additional support that these identified major QTL have the potential for further consideration to improve grain yield under a range of moisture and environmental conditions. Our results also illustrated sorghum yield during mid grain-filling and maturity stages are governed by some common genomic regions. Further, a strong association between final grain yield per panicle with grain yield per panicle at mid grain-filling stage (Fig. 3), also supports the hypothesis that grain yield during earlier stages can be used as a reliable predictor for potential final yield.

All the identified QTL for final panicle neck diameter and panicle neck diameter at the mid grain-filling stage in sorghum are reported for the first time, which adds novelty to the findings. Two out of three hotspot identified in this study contained QTL for both panicle neck diameter at mid grain-filling stage and final panicle neck diameter, suggesting common genomic regions governing panicle neck diameter at different developmental stages during grain filling. Hence, these QTL hotspots on chromosomes 1 and 2 after further validation can be valuable targets for improving source-sink dynamics and productivity in sorghum. A significant positive relationship between panicle neck diameter with grain yield per panicle and chlorophyll index suggests a key role played by source-sink-connecting panicle neck diameter as a route for assimilate transfer from source to the sink. Similar relationship has also been demonstrated previously in rice (Quinones et al. 2017).

Effective quantum yield QTL were inconsistent across environments

Four major QTL for effective quantum yield of photosystem II were detected on chromosomes 2, 6, 7 and 8 which explained 12.2, 17.4, 16.4 and 11.1% of the phenotypic variation, respectively. Major QTL on chromosome 2 (*qQY2.2*) was co-localized with earlier reported QTL for chlorophyll fluorescence (Fiedler et al. 2016; Ortiz et al. 2017), both under chilling conditions in early growth stages. This finding suggests that *qQY2.2* region could be a potential target for multiple abiotic stress tolerance at different growth stages, a hypothesis that requires additional investigation. The remaining 3 major QTL (*qQY6.1*, *qQY7.1* and *qQY8.1*) were reported for the first time in sorghum. Very few studies have reported QTL associated with chlorophyll fluorescence and gas exchange traits in field-grown sorghum except Sukumaran et al. (2016). Further, minor QTL for QY identified in this study (*qQY2.1*; *qQY3.1*, *qQY4.1*, *qQY8.2*; *qQY9.1*) co-localized with the previously reported loci from association mapping studies (Fiedler et al. 2016; Ortiz et al. 2017). Similar to the major QTL on chromosome 2, these were identified at seedling stage measured under chilling tolerance studies. However, none of these QTL were detected at more than one environment, or multiple measurements within any environment, indicating strong G×E interaction for this trait. Further, population grown in environment 5 which experienced water-limited conditions recorded significantly higher effective quantum yield of photosystem II, compared to other environments (Supplementary Table 2g). Although leaf area was not recorded in our experiments, drought stress is shown to decrease leaf area and consequently net photosynthesis on whole plant basis despite having higher photosynthesis rate (Garrity et al. 1984). In addition, effective quantum yield of photosystem II was not consistently associated with any other traits in the study suggesting narrow implications of using this trait alone in assessing or improving sorghum productivity.

Most chlorophyll index QTL were not co-localized with stay-green QTL

SC35 is a stay-green post-flowering drought stress tolerant line. Four major QTL (*Stg1*, *Stg2*, *Stg3* and *Stg4*) associated with stay-green have been consistently identified in BTx642 background in several studies (Tuinstra et al. 1997; Crasta et al. 1999; Subudhi et al. 2000; Xu et al. 2000; Sanchez et al. 2002). Stay green rating and chlorophyll index are highly correlated under post-flowering drought stress and chlorophyll index QTL are shown to co-localize with stay-green QTL (Crasta et al. 1999; Xu et al. 2000). Favorable alleles for chlorophyll index and stay-green QTL in these studies were contributed by stay-green lines. The source

of stay green is a durra landrace, IS12555 for both SC35 and BTx642. However, only one stay-green QTL (*Stg4*) overlapped with chlorophyll index QTL identified in this study. Other three major stay-green QTL (*Stg1*, *Stg2*, *Stg3*) were not detected in the study. Another parent in this study (RTx430) is a non-senescent line and hence maintains similar chlorophyll index as a stay-green line under non-water-limiting conditions (Miller 1984; Crasta et al. 1999), which may be one of the reasons for lack of detection of all stay-green QTL in this study. In addition, all the stay-green QTL identified previously in B35 × RTx430 RILs were under post-flowering drought stress conditions (Crasta et al. 1999). The study reported chlorophyll index of stay-green line (B35) was higher than RTx430 under post-flowering drought environment but not under irrigated conditions. In this study, except under the natural dryland environment in Hays (environment 5), none of the other environments were exposed to the level of water-limited conditions needed to identify stay-green QTL. Furthermore, chlorophyll index QTL (*qCh5.2*) co-localized with *Stg4* was detected only in environment 5 providing further evidence that post-flowering water-limited conditions is necessary to identify stay-green QTL in sorghum. Previous studies have reported that the strength of identifying stay-green loci and their association with grain yield would diminish as conditions shift towards non-water limiting environments (Tuinstra et al. 1997; Jordan et al. 2012).

Transgressive segregation played a key role in QTL identification

Despite no significant differences between the parents in most of the environments, major QTL were identified for final grain yield, grain yield at mid grain-filling stage, effective quantum yield of photosystem II and chlorophyll index. Transgressive segregation was observed on both ends (Fig. 1) for all studied traits which suggests both the parents possessed favorable alleles for the measured traits. Favorable allele in detected QTL for all traits were contributed by both parents which further support favorable alleles distribution between parents. Similar levels of transgressive segregation in sorghum with or without differences between the parents have been reported in several previous studies (Rami et al. 1998; Ritter et al. 2008; Reddy et al. 2013, 2014; Gelli et al. 2016; Sukumaran et al. 2016).

Conclusions

The results demonstrated a strong positive phenotypic and genotypic association between panicle neck diameter and grain yield per panicle, indicating the possibility of targeting a common QTL to improve both traits simultaneously

through MAS. In addition to the identified major QTL for each trait, novel QTL for effective quantum yield, panicle neck diameter and chlorophyll index in sorghum would add to existing knowledge and provides additional opportunities for sorghum improvement. Three QTL hotspots identified from this study could be important resources for MAS and breeding for simultaneous improvement of multiple traits. These key genomic regions, after validation have the potential for pyramiding favorable alleles for improving source–sink dynamics in sorghum to improve its productivity under diverse environments.

Author contribution statement AC, RP and SVKJ designed the experiments; AC executed the experiments. AC, MP, DS, RB, TO and HO collected the data; HP performed QTL analysis; LF, PK and WLN validated the QTL analysis; AC, RP, HP, KM and SVKJ interpreted results and drafted the manuscript. All authors contributed towards editing and finalizing the manuscript.

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Declarations

Conflict of interest Authors declare no conflict of interest.

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